MIL–I–46574A(MU)
29 NOVEMBER 1963
SUPERSEDING
MIL–I–46574(ORD)
15 SEPTEMBER 1961

MILITARY SPECIFICATION

IGNITER, PERCUSSION-ELECTRIC, M30 LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling, and packing for one type of igniter designated as Igniter, Percussion Electric, M30, for use in artillery primers.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–A–2550 —Ammunition and Special Weapons, General Specification for.
MIL–I–45208 —Inspection Requirements, General Specification for.
MIL–I–45607 —Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL–STD–105D—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–109 —Quality Assurance Terms and Definitions.
MIL–STD–650 —Explosive: Sampling, Inspection and Testing.
MIL–STD–1235 —Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

7548484 —Igniter, Percussion-Electric, M30, (For Artillery Primers) Loading Assembly.
8796692 —Box, Packing, Ammunition for Igniter, Percussion-Electric M30. Assembly, Details, Packing and Marking.
8796693 —Box, Set-Up, Packing Ammunition, for Igniter, Percussion, Electric M30. Assembly, Details, Packing and Marking.

PUBLICATIONS

ORDNANCE CORPS

IEL–7548484 —Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.** Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 **Igniters.** The igniters shall comply with

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all requirements specified on Drawing (dwg.) 7548484, and with all requirements specified in applicable specifications.

3.3 Moisture content of ignition charge composition. The moisture content of the ignition charge composition at the loading station at the time of loading the ignition element shall not exceed 0.10 percent when tested as specified in 4.3.1 (see 6.4).

3.4 Electrical resistance. The igniters shall have an electrical resistance of not less than 0.80 ohm nor more than 1.80 ohms when tested as specified in 4.3.2.

3.5 Functioning

3.5.1 *Electrical.* The igniters shall function within .2 seconds maximum (max.) when energized with a direct current of 1.25 amperes, max. Testing shall be as specified in 4.3.3.1.

3.5.2 *Percussion.* The igniters shall function when tested as specified in 4.3.3.2.

3.6 Non-functioning

3.6.1 *Electrical.* The igniters shall not function when energized with a direct current of 190 to 200 milliamperes when tested as specified in 4.3.4.1.

3.6.2 *Percussion.* The igniters shall not function when tested as specified in 4.3.4.2.

3.7 Workmanship. The parts shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
(b) Number of units of product inspected.
(c) Results obtained for all inspection performed.
(d) Drawing, specification number and date, together with an identification and date of changes.
(e) Certificates of analysis on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
(f) Number of items in the lot.
(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the

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agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions

4.2.1 *Lot formation.* A lot shall consist of igniters produced by one manufacturer in one unchanged process, in accordance with the same drawing, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot interfix number. Product shall be submitted for inspection in accordance with Standard MIL–STD–105D (or Standard MIL–STD–1235, when applicable). Each lot shall contain:

(a) Metal parts of one lot interfix number from one manufacturer.
(b) Ignition charge composition of not more than one lot from one manufacturer.
(c) Ignition elements of not more than one lot interfix number from one manufacturer.
(d) Percussion primers of one interfix number from one manufacturer.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Ignition element assembly (see dwg. 7548479 covering a detail of dwg. 7548484).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Plunger pin missing	Visual	01001
Major:	AQL 0.25 percent		
101.	Ignition charge missing, broken or chipped	Visual	01002
Minor:	AQL 1.50 percent		
201.	Total length, max.	Gage	01003
202.	Ignition charge above metal surface	Visual Straight edge	01004
203.	Ignition charge on exterior surface	Visual	01005
204.	Evidence of poor workmanship (see 3.7)	Visual	01006

4.2.2.2 Assembly, prior to inserting closing plug (see dwg. 7548484).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined		

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Categories	Defects	Method of inspection	Code No.
Major:	AQL 0.25 percent		
101.	Closing disc missing or mutilated	Visual	02001
102.	More than one closing disc	Visual-Manual	02002
Minor:	None defined		

4.2.2.3 Assembly (see dwg. 7548484).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Closing plug not fully seated or loose to finger pressure	Manual	03001
102.	Closing plug missing	Visual	03002
103.	Flash hole obstructed	Visual	03003
104.	Plunger or plunger insulator protrudes beyond base	Visual	03004
Minor:	AQL 1.50 percent		
201.	Plug stake missing	Visual	03005
202.	Thread or flange damaged	Visual	03006
203.	Evidence of poor workmanship (see 3.7)	Visual	03007
204.	Stamping or marking misleading or unidentifiable (as applicable)	Visual	03008

4.2.2.4 Box, set-up, packing (see dwg. 8796693).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Assembly damaged so that contents are exposed or liable to become exposed	Visual	04001
102.	Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal the box completely	Visual	04002
Minor:	AQL 1.00 percent		
201.	Contents move when box is shaken	Manual	04003
202.	Carton torn, cut or punctured	Visual	04004

4.2.2.5 Sealed barrier bag (see dwg. 8796693).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Seal missing, incomplete or incorrectly located	Visual	05001
102.	Bag punctured, torn or cut	Visual	05002
Minor:	None defined		

4.2.2.6 Box, packing (see dwg. 8796692).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Board broken or split	Visual	06001

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Categories	Defects	Method of inspection	Code No.
102.	Contents exposed	Visual	06002
103.	Strapping missing, broken or loose	Visual-Manual	06003
Minor:	AQL 1.50 percent		
201.	Marking misleading or unidentifiable	Visual	06004
202.	Strapping improperly assembled	Visual-Manual	06005
203.	Contents move when box is tilted	Manual	06006

4.2.3 *Testing.*

4.2.3.1 Moisture content of ignition charge composition (see 3.3) Major defect—Code No. 07001. The contractor shall test for verification at least one sample of approximately ten grams of ignition charge composition during each eight hours production of igniters. Composite samples shall not be used in determining compliance. If the moisture content is in excess of the specified requirement and loading has not begun, the lot of composition shall be rejected. If loading has begun, the igniters loaded with the non-conforming composition shall be rejected. The determination shall be performed as specified in 4.3.1.

4.2.3.2 Electrical resistance (see 3.4) Major defect—Code No. 03009. This test shall be conducted 100 percent. Any igniter which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.3.2, using equipment in accordance with 4.2.4.

4.2.3.3 Functioning (see 3.5).

4.2.3.3.1 Electrical—Major defect—Code No. 03010. Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified two hundred and fifty samples shall be randomly selected from each lot for this test. If three or more igniters fail to comply with the applicable requirement, the lot shall be rejected. If one or two defects are found, a second sample of five hundred igniters shall be tested. If the combined number of defectives in both the first and second samples is three or more, the lot shall be rejected. The test shall be performed as specified in 4.3.3.1 using equipment in accordance with 4.2.4.

4.2.3.3.2 After three consecutive lots have met the criteria of 4.2.3.3.1, a sample of 125 igniters shall be selected from each lot for this test. If two or more igniters fail to comply with the applicable requirement, the lot shall be rejected. If only one defect is found, a second sample of 250 igniters shall be tested. If the combined number of defectives in both the first and second samples is two or more, the lot shall be rejected. The test shall be performed as specified in 4.3.3.1 using equipment in accordance with 4.2.4.

4.2.3.3.3 Percussion—Major defect—Code No. 03011. Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, two hundred and fifty samples shall be randomly selected from each lot for this test. If three or more igniters fail to comply with the applicable requirement, the lot shall be rejected. If one or two defects are found, a second sample of five hundred igniters shall be tested. If the combined number of defectives in both the first and second samples is three or more, the lot shall be rejected. The test shall be performed as specified in 4.3.3.2 using equipment in accordance with 4.2.4.

4.2.3.3.4 After three consecutive lots have met the criteria of 4.2.3.3.3, a sample of 125 igniters shall be selected from each lot for this test. If two or more igniters fail to comply with the applicable requirement, the lot shall be rejected. If only one defect is found, a second sample of 250 igniters shall be tested. If the combined number of defectives in both the first and second sample is two or

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more, the lot shall be rejected. The test shall be performed as specified in 4.3.3.2 using equipment in accordance with 4.2.4.

4.2.3.4 Non-functioning (see 3.6).

4.2.3.4.1 Electrical—Critical defect—Code No. 03012. Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, one hundred and fifty samples shall be randomly selected from each lot and shall be subjected to the test specified in 4.3.4.1 using equipment in accordance with 4.2.4. If any failure occurs, the lot shall be rejected. Igniters subjected to this test shall be destroyed.

4.2.3.4.2 After three consecutive lots have met the criteria of 4.2.3.4.1, a sample of 50 igniters shall be randomly selected from each lot for this test. If one or more igniters fail to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.4.1 using equipment in accordance with 4.2.4. Igniters subjected to this test shall be destroyed.

4.2.3.4.3 Percussion—Critical defect—Code No. 03013. Beginning with the first lot produced and continuing until three consecutive lots complied with the applicable requirements specified, one hundred and fifty samples shall be randomly selected from each lot and shall be subjected to the test specified in 4.3.4.2 using equipment in accordance with 4.2.4. If any failure occurs, the lot shall be rejected. Igniters subjected to this test shall be destroyed.

4.2.3.4.4 After three consecutive lots have met the criteria of 4.2.3.4.3, a sample of 50 igniters shall be randomly selected from each lot for this test. If one or more igniters fail to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.4.2 using equipment in accordance with 4.2.4. Igniters subjected to this test shall be destroyed.

4.2.3.5 Determination of ignition charge composition (see dwg. 7548480) Major defect —Code No. 07002. A sufficient amount of ignition charge composition shall be selected from each lot for analysis. If the composition fails to comply with the applicable requirements and loading has not begun, the lot of composition shall be rejected. If loading has begun, the igniters loaded with the non-conforming composition shall be rejected. The determination shall be performed as specified in 4.3.5.

4.2.4 *Inspection Equipment.* Index of Inspection Equipment Number IEL 7548484 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and wherever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 Determination of moisture content of ignition charge composition.

4.3.1.1 *Special apparatus*

(a) Distillation receiver of borosilicate glass. This item may be conveniently fabricated by attaching a

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standard taper No. 2 stop-cock to a crankcase dilution receiver ASTM Standard D–322 and drawing-out and curving the outward stem to serve as a draw-off spigot.

(b) A water cooled condenser protected by a drying tube containing a suitable desiccant.

(c) Apparatus in accordance with Standard MIL–STD–640 method 101.4.

(d) Electric hot plate, adjustable, with triethylene glycol baths on each heater.

4.3.1.2 *Special material*

(a) Materials in accordance with Standard MIL–STD–650 method 101.4.

(b) Trichlorethylene, Dupont Triclene or equivalent, prepared not longer than 48 hours previous to use by shaking with water in a separatory funnel at least two minutes and left to stand in contact with water at least overnight.

4.3.1.3 *Procedure.* An accurately weighed sample of approximately 5 grams shall be placed in a dry boiling—flask which is immediately re-stoppered. (Note: In the event that the sample has begun to harden before weighing, it should be rubbed gently through a 20 mesh screen with a rubber stopper. Just previous to use the condenser and receiver shall be rinsed with anhydrous methanol which is discarded. (A small residue of methanol in the condenser and receiver does no harm). The stopper shall be removed from the flask, the receiver (its stopcock closed) shall be inserted in its place, and a 100 milliliter portion of trichlorethylene saturated with water shall be pipetted into the receiver so that the column remains substantially filled. The assembly shall be placed in the bath on the hot plate and the condenser inserted into the receiver. The temperature of the bath shall be adjusted so that the reflux from the condenser shall be about 2 drops per second and the distillation shall be continued for 2 hours, at the end of which time the bath shall be allowed to cool. When condensate no longer drips from the receiver into the flask the contents of the receiver shall be drawn off through the stopcock into a dried 50 milliliter volumetric flask. The drying tuble shall be removed and the condenser and receiver washed with a few 5 milliliter portions of anhydrous methanol which shall be drained into the volumetric flask. The boiling-flask and condenser shall be detached from the receiver and the receiver further washed with successive portions of methanol until the volumetric flask is filled to the mark. The volumetric flask shall be stoppered and mixed by shaking. A blank shall be run using the above procedure. The flash may be stoppered and set aside until titrated. Volumetric flask shall be emptied into a dried 300 milliliter "tallform" beaker containing a magnetic stirring bar, and the flask shall be washed out by filling it once again to the mark with anhydrous methanol, shaking, and pouring it into the beaker. The beaker shall be quickly mounted on a Karl Fischer titrating stand in accordance with Standard MIL–STD–650 method 101.4. The solution shall be titrated in accordance with Standard MIL–STD–650 method 101.4 beginning with paragraph 5.9, except that a bracket shall be in the equation after the letter F and before 100 on page 4 of method 101.4 of Standard MIL–STD–650.

4.3.2 *Electrical resistance.* The igniters shall be tested for electrical resistance by applying a direct current of 20 milliamperes, max., using the plunger as one contact point and the igniter body as the other. Observation shall be made for compliance with the applicable requirement.

4.3.3 *Functioning.*

4.3.3.1 *Electrical.* The igniters shall be tested for electrical functioning by applying a direct current of 1.25 amperes, max., using the plunger for one contact point and the igniter body as the other. Observation shall be made for functioning within .2 seconds, max.

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4.3.3.2 *Percussion.* The igniters shall be inserted in a holder and the holder placed in the base of the primer test fixture (see 4.2.4). The firing pin holder with the firing pin shall be inserted into the weight cup support cover so that the firing pin holder firmly contacts the upper face of the holder. The height of drop of a standard 1½-inch diameter steel ball weighing approximately 7.94 ounces shall be adjusted to 20 inches, max. The ball shall be released and observation made for functioning.

4.3.4 *Non-functioning*

4.3.4.1 *Electrical.* The igniter shall be tested for electrical functioning by applying a direct current of 190 to 200 milliamperes, using the plunger for one contact point and the igniter body as the other. The time of the current flow shall be one second minimum. Observation shall be made for non-functioning.

4.3.4.2 *Percussion.* The test shall be performed as specified in 4.3.3.2, using the same weight steel ball, except that the height of drop shall be adjusted to 6 inches, minimum. The height of the drop shall be measured as the height from the top of the firing pin to the bottom of the ball. The ball shall be released and observation made for non-functioning.

4.3.5 *Determination of ignition charge composition mixture (see 6.6).* The chemical composition of the ignition charge shall be considered satisfactory if it is in accordance with the following:

Component	*Percentage*
Potassium Chlorate	44.5 plus or minus 2
Lead Thiocyanate	35.5 plus or minus 2
Charcoal	20.0 plus or minus 1.5
Lacquer Solids (added)	1.6 plus or minus 0.2

4.3.5.1 *Lacquer solids.* Approximately 1.5 grams of the sample shall be transferred to a tared medium porosity filtering crucible and allowed to air dry for about an hour. The crucible and contents shall be placed in a vacuum oven for two hours maintained at a temperature of 55 plus or minus 2 degrees Centigrade (C.). The crucible and contents shall be cooled in a desiccator and weighed. The procedure shall be repeated until the change in weight does not exceed 0.5 milligrams. The contents of the crucible shall be extracted with ten 25 milliliter (ml.) portions of acetone saturated with potassium chlorate and lead thiocyanate. The crucible and contents shall be aspirated until the odor of acetone is no longer perceptible and then dried in a vacuum oven maintained at 55 plus or minus 2 degrees C. for two hours. The crucible and contents shall be cooled in a desiccator and weighed. The percent lacquer solids shall be calculated as follows:

Percent lacquer solids = 100
A minus 100 B divided by B.

where:

A = the weight of the sample dried in the oven

B = the weight of dried sample after acetone extraction.

4.3.5.2 *Charcoal.* Approximately 1.5 grams of the sample shall be transferred to a tared medium porosity filtering crucible. The contents of the crucible shall be extracted with ten 25 ml. portions of acetone saturated with potassium chlorate and lead thiocyanate. The crucible and contents shall be aspirated until the odor of acetone is no longer perceptible and then dried in a vacuum oven maintained at 55 plus or minus 2 degrees C. for two hours. The crucible and contents shall be cooled in a desiccator and weighed. Using a Fisher high form filtrater or equivalent extractor the sample shall be extracted with ten 25 ml. portions of a 5 percent potassium nitrate solution followed with 50 ml. of distilled water. A glass rod shall be used to break up any lumps in the sample during the extraction process, the filtrate shall be reserved for the determination of lead thiocyanate. The crucible and contents shall be aspirated for several minutes then placed in an oven maintained at 100 plus or minus 5 degrees C. for two hours. The crucible and contents shall be cooled in a desiccator and weighed. The percent charcoal shall be calculated as follows:

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Percent charcoal = 100 A divided by B

where:

A = the weight of sample after acetone and 5 percent potassium nitrate solution extraction.

B = the weight of sample after acetone extraction.

4.3.5.3 *Lead thiocyanate.* Exactly 35 ml. of standardized 0.1 normal (N) silver nitrate shall be added to the filtrate reserved from the charcoal determination. The flask and contents shall be swirled and 100 ml. of 3N nitric acid (nitric acid used should be water-white and free of oxides of nitrogen) shall be added. The swirling of the mixture shall be continued for five minutes, 5 ml. of saturated solution of ferric ammonium sulfate shall be added and the solution titrated with standardized 0.1 N potassium thiocyanate solution of the first permanent pale rose coloration. The percentage of lead thiocyanate in the sample shall be calculated on a dry basis as follows:

Percent lead thiocyanate = 16.17
AE minus 16.17 CD divided by B.

where:

A = volume of silver nitrate solution used.

B = weight of the sample after acetone extraction (see 4.3.5.2).

C = volume of potassium thiocyanate solution used.

D = normality of potassium thiocyanate solution.

E = normality of the silver nitrate solution.

4.3.5.4 *Potassium chlorate.* The percent of potassium chlorate shall be determined by subtracting the total percentage of charcoal and lead thiocyanate from one hundred percent.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level C.* Packaging shall be as specified on Drawing 8796693.

5.2 Packing.

5.2.1 *Level C.* Packing shall be as specified on Drawing 8799692.

6.3 Marking. The boxes shall be marked in accordance with Drawing 8796692.

5.4 Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 *Inspection Equipment Lists.* Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection Equipment List codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in

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the Inspection Equipment List by the following codes:

CDOF —Contractor's design responsibility on Ordnance format; ORDM–608–12.
CDCF —Contractor's design responsibility on contractor format.
OD —Ordnance design
ODMU —Ordnance design, mandatory for use.
CE —Commercial equipment.
SCD —Specification Control Drawing.

6.3.3 *Ordnance designs.* Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and non-mandatory. When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawings.* Specification control drawings depict the minimum requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government . Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608–12, and, in addition, contain the detail and information specified above.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

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6.4 Loading of charge. Although the charge is part of the M4 Ignition Element, it is not applied until the element is ready to be assembled into the M30 Igniter (see dwg. 754-8479).

6.5 Check tests for possible deterioration (applicable to igniter from storage). If the total elapsed time between original acceptance of an igniter lot and assembly of that lot into the ammunition components with which it is to be issued exceeds two years, or if the igniters have been subjected to adverse conditions at any time since previous tests, the igniter lots shall be subjected to, and must satisfactorily pass, the following check test immediately before the lot is used.

6.5.1 *Functioning.*

6.5.1.1 *Electrical.* The sampling plan for this test shall be in accordance with Table I. Any igniter which fails to comply with the applicable requirement shall be classed defective. The test shall be performed as specified in 4.3.3.1.

6.5.1.2 *Percussion.* The sampling plan for this test shall be in accordance with Table I. Any igniter which fails to comply with the applicable requirement shall be classed defective. The test shall be performed as specified in 4.3.3.2.

6.5.2 *Non-functioning.*

6.5.2.1 *Electrical.* The sampling plan for this test shall be specified in 4.2.3.4.2, with the method of testing in accordance with 4.3.4.1.

TABLE I

Sample	Sample		Combined samples	
	Size	Size	Acceptance number	Rejection number
First	75	75	*	2
Second	75	150	0	3
Third	75	225	0	3
Fourth	75	300	1	4
Fifth	75	375	2	5
Sixth	75	450	2	5
Seventh	75	525	4	5

* No acceptance on this sample

6.5.2.2 *Percussion.* The sampling plan for this test shall be as specified in 4.2.3.4.4, with the method of testing in accordance with 4.3.4.2.

6.5.3 *Electrical resistance.* Each igniter shall be tested as specified in 4.3.2. If any igniter fails to comply with the applicable requirement, it shall be removed from the lot.

6.6 Determination of ignition charge composition. During the manufacture of the ignition charge composition up to two parts of solvent are added to one part of Egyptian lacquer in order to facilitate the blending of the mixture. Because of the volatility of the solvent, the solvent content of the ignition charge composition is found to be variable. Because of this the determination of the potassium chlorate, lead thiocyanate and charcoal content are determined on a lacquer free basis and the lacquer solids on added basis.

Custodian:
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Preparing activity:
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Project No. 1390–A328

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